

From Static Systems to Unending Flow: Overcoming Gödel's Incompleteness through Alexandre de Castro Prado's Unending-Logical Actualism

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ABSTRACT

This paper proposes a critical reassessment of Kurt Gödel's Incompleteness Theorems (1931) through the lens of Unending-Logical Actualism, a philosophical system developed by Alexandre de Castro Prado. While Gödelian logic demonstrates the existence of unprovable statements within finite formal systems, establishing an insurmountable limit for pure reason, Prado's thesis contends that such incompleteness stems from a static and atemporal conception of logic. By introducing dynamic variables—such as the update operator ($\oplus$), the free-will vector (α), and the notion of the Productive Unknown (?)—it is demonstrated that what Gödel classifies as unprovable actually constitutes a processing latency within an open system. The article stratifies truth into three levels (Evident, Probable, and Possible) and presents the logical formalization that incompleteness is destroyed by the being's power of updating. It concludes that the universe is not a flawed or lacunar system, but an "Unending" structure, where knowledge is a continuous process of converting potency into act. Reality is rescued as an uninterrupted flow where Gödel's uncertainty is transmuted into the hope of perpetual discovery, founding a new logic that integrates ethics, science, and presence.

KEY-WORDS

Actualism; Incompleteness; Unending; Dynamic Logic; Alexandre de Castro Prado.

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INTRODUCTION

The history of Western logic is marked by moments of closure and openness. In 1931, by publishing his Incompleteness Theorems, Kurt Gödel seemed to have sealed the fate of reason within a glass bell jar: if a system is consistent, it is necessarily incapable of self-completion. However, what the analytical tradition accepted as an insuperable limit is, from the perspective of Alexandre de Castro Prado, the symptom of a static ontology. Unending-Logical Actualism emerges not to deny Gödel's mathematical rigor, but to rescue it from its immobility, reinserting logic into the uninterrupted flow of being. The fundamental cornerstone of this surpassing lies in the redefinition of the concept of Truth ($\mathbb{V}$). Where classical logic operates within a restricted binarism (true/false) or a purely syntactic provability, Prado establishes a hierarchy of degrees of access to reality, formalized by the Axiom of Epistemic Stratification:

$$\mathbb{V} = [\mathbf{EV}, \mathbf{PrV}, \mathbf{PoV}]$$

Wherein:

EV (Evident Truth): Refers to the direct presence of the object to consciousness (scientific or factual evidence), in which being attests to reality without the need for syllogistic mediation. It is the anchoring point of the Actual (A).

PrV (Probable Truth): The field of philosophy, formal logic, and mathematics, where truth depends on a chain of demonstration ($\vdash$). It is here that Gödel's analysis came to a standstill.

PoV (Possible Truth): The dimension of the Infinite (I), inhabited by that which has not yet been actualized but possesses the potency of being. It is the "moral truth" and the horizon of discovery.

By recognizing that Gödel's "unprovable" merely inhabits the PoV stratum awaiting transition to EV, Prado destroys the perception of systemic "failure." Incompleteness ceases to be a wall and becomes the very condition of possibility for a living and inexhaustible universe.

To demonstrate the robustness of this thesis, the present work will proceed according to the following roadmap: first, we will examine Gödel's logical *iter* and his conclusion on incompleteness. Next, we will present Actualism in its "Light Version," demonstrating how the introduction of the temporal variable and processing latency already effects a technical surpassing of the Gödelian limit. Immediately following, we will address potential logical objections to Prado's system, followed by their respective refutations. Expanding the scope, we will deal with the psychological prejudices and the contemporary resistance to the classical metaphysics upon which Prado relies (Act, Potency, Finality), refuting them forcefully. Finally, we will apply this logic to the "Extended Version" of the theory, culminating in a conclusion that synthesizes the rescue of intellectual hope through the Productive Incognita.

DEVELOPMENT

A) THE GENESIS OF INCOMPLETENESS: KURT GÖDEL'S LOGICAL ITER

To understand the surpassing proposed by Actualism, it is imperative to reconstruct, with academic rigor, the edifice erected by Gödel in 1931. Gödel's central problem was to respond to Hilbert's Program, which sought to prove that mathematics was complete (every truth has a proof) and consistent (there are no contradictions).

Hilbert's Program (David Hilbert, 1920s) was the quest for the "Theory of Everything" in mathematics, based on three pillars that can be expressed in the following logical notation:

I — Hilbert's Program (H)

Let **S** be a formal axiomatic system (such as arithmetic); Hilbert's goal was to prove that:

Consistency (Consis S)

The system cannot generate contradictions. One cannot derive a formula **A** and its negation $\neg A$.

Notation:

$$\neg(S \vdash A \wedge \neg A)$$

Meaning: "It is impossible for S to demonstrate A and not-A ($\neg A$) simultaneously."

Completeness (Comp S)

The system must be able to decide on any well-formed statement within it.

Notation:

$$\forall A \in S, (S \vdash A \vee S \vdash \neg A)$$

Meaning: "For any sentence A, either the system proves A, or the system proves the negation of A." (There is no "limbo" or unprovable truths).

Decidability (Decid S)

There must exist an algorithm (a finite mechanical method) that determines whether any sentence is or is not a theorem.

Notation:

$$\exists \text{ Algorithm } \mathcal{M} : \mathcal{M}(A) \rightarrow [1,0]$$

Meaning: "There exists a method that, for any sentence A, answers 'yes' (1) or 'no' (0) within a finite time."

II — Gödel's Incompleteness

Gödel demonstrated the opposite through a process of four fundamental stages:

The Arithmetization of Syntax (Gödel Numbering)

The first step was to translate logic into the language of numbers. Gödel established an injective function that assigns a unique natural number (**g**) to every symbol, formula, and sequence of formulas (proof).

Notation:

$$G(S) = n$$

Meaning: Through this encoding, speaking about "proofs" became equivalent to performing arithmetic calculations. The system gained a "mirror image" of itself within its own structure.

The Provability Predicate

Gödel formalized the relationship between a sequence of logical steps and a conclusion.

Notation:

$$Dem(x,y)$$

Meaning: "The sequence of formulas encoded by number **x** is a demonstration of the formula encoded by number **y**."

Implication: If the system can find such an **x** for a given **y**, then **y** is provable ($\vdash y$).

The Diagonal Lemma (Self-Reference)

Using a refined mathematical technique, Gödel constructed a special sentence, which we shall call Sentence G, that makes an assertion about its own Gödel number (**g**).

Notation:

$$G \leftrightarrow \neg \exists x \text{ Dem}(x, g)$$

Meaning: Sentence G asserts: "There is no number **x** that is the proof of **g** (myself)."

The Paradox: Gödel created a mathematical sentence that explicitly says: "I am not provable."

The Inevitable Conclusion: Incompleteness

By analyzing the truth of G, Gödel arrives at a logical checkmate:

- If the system proves G, then G is false (as it claims not to be provable), which would make the system inconsistent.
- If the system is consistent, it cannot prove G. However, if it does not prove G, then what G says is true.

Final Notation:

$$S \not\vdash G \mid G \in \mathcal{V}$$

Reading: "System S fails to demonstrate G, even though G is a Truth within the model of natural numbers."

Grounding the "Wall":

Gödel's conclusion establishes that Truth is a much larger ocean than Provability. For Gödel and 20th-century logic, this meant that mathematics was condemned to be eternally "gappy." The system is static; it is what it is, and its limitations are concrete walls insuperable by pure logic.

Technical Commentary:

In this section, we establish that Gödel views the system as a photograph. If the photo is incomplete, it will be incomplete forever. This is the "statism" that Alexandre de Castro Prado will identify as the original error.

The Contrast with Gödel and Prado's Foreshadowing

Hilbert firmly believed in the motto: "*Wir müssen wissen, wir werden wissen*" (We must know, we will know).

By publishing his results, Gödel annihilated both Completeness and Consistency (proving that a system cannot prove its own consistency).

In Gödel's view, Hilbert's failure was a final defeat. However, for Alexandre de Castro Prado, Hilbert's mistake was not his optimism ("we will know"), but the attempt to enclose that knowledge within a static system.

While Hilbert wanted the system to be complete *now*, Prado will show that the system is Unending (*Infindo*)—completing itself through time.

B) THE SYSTEMIC RESPONSE: ACTUALISM IN ITS LIGHT THEORY

For item (B), we present the first level of overcoming proposed by Alexandre de Castro Prado. Here, the focus is technical: the transition from a Logic of Static Sets to a Logic of Dynamic Systems.

Alexandre de Castro Prado's initial overcoming lies in the denunciation that Gödel committed a dimensional category error: he treated Knowledge as a fixed geometric object, when it is, in essence, a procedural event. The "Light Theory" introduces system physics into formal logic through three pillars:

The Redefinition of the System as a Temporal Function

While for classical logic system S is an immutable set of axioms, Prado proposes that the system is an entity in flux.

Notation:

$$S = f(t)$$

Meaning: The state of knowledge (S) is dependent on the time variable (t). What is true within the system is not a constant, but a value that expands chronologically.

The Concept of Processing Latency (Δt)

Alexandre reinterprets Gödel's "Undemonstrable." If Gödel found a sentence G that cannot be proven in S , Prado does not see this as a system failure, but a processing delay.

Notation:

Incompleteness $\equiv (G \in \text{Buffer})$ where $T_{\text{processing}} > t_{\text{current}}$

Meaning: Sentence G is not "unprovable"; it is merely latent. It is in a systemic waiting queue. Incompleteness is simply the time required (Δt) for raw reality to be "chewed" by logic.

The Systemic Upgrade Operator ($\oplus$)

Unlike the static union of sets, Prado introduces a dynamic integration operator. The system does not merely "contain" data; it assimilates it.

Notation:

$$S_{t+1} = S_t \oplus \Delta [I(t)]$$

How to read: "The system at the following instant (S_{t+1}) is equal to the current system (S_t) integrated ($\oplus$) by the update (Δ) of the potential that was waiting in the Unending (I)."

The Dynamic Conception of Chaos

At this level, the theory establishes that what we call Chaos (C) possesses no negative ontological reality; it is merely Unprocessed Order.

Light Axiom of Chaos:

$$C(t) \in I(t) \rightarrow \exists t + n \mid C \rightarrow O$$

Meaning: What appears confusing or undemonstrable today is just an Order (O) that still inhabits the Unending (I) and will inevitably be updated by the system at a future time $t+n$.

Technical Commentary:

In Light Theory, Alexandre de Castro Prado replaces Gödel's "No" with "Not yet." He proves that Gödel's mathematics is a special case of a system that was arbitrarily frozen in time. By unfreezing the system, Incompleteness disappears and gives way to Unendingness.

To conclude item (B) with rigor, let us formalize the clash between Gödel's statics and Prado's dynamics. This "debate" is not just about ideas, but incompatible logical structures: one that treats truth as an object (paper) and another that treats it as a process (execution).

ADDENDUM TO ITEM B: NOTATIONAL CLASH - "PAPER MATHEMATICS" VS. "SYSTEM IN EXECUTION"

Gödel's Thesis: The System as a File (Paper)

In Gödel's logic, knowledge is a closed set of axioms Σ . If the truth G is not contained within Σ , it is "outside" forever.

Axiom of Statics:

$$\Sigma = [a_1, a_2, \dots, a_n]$$

The Failure Condition:

$$\exists G : (G \models V \wedge G \notin \Sigma) \rightarrow S = \text{Incomplete}$$

Meaning: If the sentence G is true but is not written on the "paper" (within set Σ), the system is broken. It is a logic of inventory (stock).

Prado's Antithesis: The System as Flux (Execution)

In Alexandre de Castro Prado's "Light Theory," knowledge is not a file (Σ), but a Processing (Δ). What Gödel calls "being outside," Prado calls "being in transit."

Axiom of Dynamics:

$$S(t) \text{ ---}\Delta\text{---} S(t + 1)$$

The Dissolution of Failure:

$$(G \notin S_t) (\nrightarrow) S = \text{Incomplete}$$

The New Condition:

$$(G \notin S_t) \rightarrow G \in \text{Queue(I)}$$

Meaning: The fact that G is not in the system at time t does not prove incompleteness, but proves latency. G is in the "Queue" of the Unending (I), awaiting the next execution cycle.

Final Confrontation in Logical Notation

Category	Gödel	Alexandre
Nature of S	$S = \text{constant}$	$S = f(t)$
Status of G	$G = \text{system error}$	$G = \text{Latent Data}$
Proof Relation	Proof $\equiv$ Being on Paper	Proof $\equiv$ Act of Updating
Conclusion	Absolute Limit	Unendingness

The Breaking Point:

Gödel works with the Negation of Proof ($\neg\text{Prov}$), while Prado works with the Differential of Proof ($dS \mid dt$).

In paper mathematics, if the proof does not exist now, it will never exist. In a system in execution, the proof is the result of the system's own movement. Alexandre proves that Gödel did not hit a wall, but rather discovered the response time of reality.

Transition Commentary:

Through this notational debate, we demonstrate that "Incompleteness" is a property only of systems that lack the time variable. By inserting t , Gödel's system "melts" and flows into Actualism.

C) THE SUBSTITUTION OPERATION: HOW ACTUALISM OVERCOMES GÖDEL IN THE "LIGHT VERSION"

In this section, we will perform the logical operation of substitution. The goal is to demonstrate that Gödel's conclusion is not a final verdict, but an isolated case within a larger and more powerful mechanics.

For the overcoming to be definitive, we must show that Gödel's Sentence (G), when inserted into Alexandre de Castro Prado's processor, ceases to generate a system error (Crash) and begins to generate an expansion cycle.

The Deconstruction of Static Negation

Gödel bases his conclusion on the sentence:

$$G \leftrightarrow \neg \text{Prov}(g)$$

In Light Actualism, the negation operator ($\neg$) is replaced by the Temporal Deferral Operator (Δt).

Gödel Notation:

$$G \equiv \text{Unprovable}$$

Prado Substitution:

$$G \equiv \text{Processing}(t)$$

Demonstration: In Prado's logic, "not being proven" is not an eternal state, but a coordinate on the time axis. Thus, Gödel's sentence does not say "I am impossible," but "My proof requires a time $t > 0$."

The Collapse of Incompleteness (Step-by-Step)

Follow the logical sequence of the overcoming:

1. **Gödel's Premise:** If $G \in V$ is true and the system is static ($S = \text{const}$), the system has failed to cover the truth.
2. **Prado's Intervention:** We introduce the Update variable ($\oplus$).

3. **Processing:** $S_{t+1} = S_t \oplus G$
4. **Resolution:** At instant $t+1$, the data that was latent ($G \in \text{Queue}$) is integrated into the set of current evidence (VE).

Result:

$$S_{t+1} \vdash G \therefore \text{Incompleteness} \rightarrow 0$$

1. Formalization of the Overcoming: The Moving Horizon Theorem

The technical overcoming does not point toward an infinite completeness (as one might erroneously assume—for reasons that will be alluded to later in this work, since, for Alexandre, the 'infinite' belongs to the divine sphere), but toward an Unending Completeness. The system does not end at an omega point; instead, it eliminates incompleteness through a flow of absorption that does not have a known—or necessary—end.

Notation of the Unending Flux:

$$\Sigma (S_t \oplus \Delta I_t) \rightarrow V_{\text{total}}$$

Meaning: "The summation (Σ) of successive system updates (S_t), integrated ($\oplus$) by the variation of potential (Δ) coming from the Unending (I), continuously converges toward the Totality of Truth (V_{total}), without ever terminating the process."

Summation (Σ): Represents the historical and continuous accumulation of knowledge.

Integrated ($\oplus$): Indicates that the new data is not merely added, but assimilated and processed by the system's structure.

Potential Variation (ΔI_t): This is the "fraction" of what was unknown (Unending) that becomes actual at time t .

Converge ($\rightarrow$): Indicates movement toward, without necessarily implying a static endpoint, respecting the Unending nature of the totality.

- Prado's Ontological Differentiation:

Knowledge of Individual Things (C_i): Possesses varying states of exhaustion:

- **$C_i = 1$:** Exhausted knowledge (full evidence).
- **$C_i \rightarrow 1$:** In the process of exhaustion.

- **Ci $\approx$ 0:** Deep latency.

Totality of Knowledge ($\forall$ total): This is the Unending structure. Unlike God (the Unique Infinite and Unreachable – whether God exists or not, as will be seen), the totality of the intelligible and material world is a moving horizon that the system pursues without ceasing.

Reformulated Reading:

While Gödel sees a structural flaw by viewing the system as something that should be "ready" (finite), Prado reveals that the system is a process of exhausting potentials. "Incompleteness" is destroyed not because we reach the Infinite, but because Prado's system never stops updating what is Unending. What Gödel called a "limit" is, in fact, the leading edge of a system that is alive and in constant "mastication" of reality.

Concept Adjustment:

With this, we clarify that the theory does not aim to reach divine omniscience (the Infinite), but rather to guarantee that, within the material and intelligible universe, there is no "wall," only a path made by walking (the Unending).

The overcoming of Gödel occurs because the system is not "trying to be complete" (in Hilbert's static sense), but is being Unending (in Prado's dynamic sense). This shift perfectly preserves the following points:

1. **The distinction between the individual and the totality:** Individual knowledge can be exhausted, while the totality remains in flux.
2. **The critique of statism:** Gödel's error remains that of judging a system in motion with the metrics of a static system.
3. **The divine frontier:** Keeps God as the unique Infinite, outside the scope of the material/intelligible logical "tool."

D) CONTESTATIONS AND REFUTATIONS: CONFRONTING ACTUALISM

In this section, we submit Unending Logical Actualism to the scrutiny of contradiction, responding to objections that attempt to uphold the validity of Gödel's

"wall." The overcoming is demonstrated at two levels: the efficiency of the autonomous system and the power of human agency.

1. The Mechanistic Layer: The Response of the Autonomous System (Without Agency)

Objection 01: "The mere introduction of time does not solve undecidability. Even a dynamic system may encounter a sentence that requires inaccessible processing time, maintaining incompleteness in practice."

Prado's Refutation: The error of this objection lies in treating the "undemonstrable" as an execution error (crash). In Actualism, the system does not interrupt the flow; it reclassifies the data through the Productive Unknown (?).

- Dynamic Labeling Notation:

$$? \in [VPo \xrightarrow{\Delta t} VE]$$

- How to read: "The Productive Unknown (?) is an element belonging to the field of Possible Truth (VPo), which is in a state of temporal transition ($\Delta t \rightarrow$) toward its conversion into Evident Truth (VE)."

- Technical Explanation: Gödel's incompleteness is a static negation ($\neg$). Prado's response is the creation of the Active Processing state. The system is complete because it possesses a logical category for the "not yet processed." The undemonstrable ceases to be a limit and becomes the fuel (raw material) for the system's next update cycle.

2. The Anthropological Layer: The Invasion of Agency (With Human Action)

Objection 02: "Logic must be a domain of pure laws. By introducing 'human action,' the system ceases to be formal and becomes dependent on the observer's subjectivity."

Prado's Refutation: Prado demonstrates that Gödel's logic is incomplete precisely because it is a "logic without a subject." Reality is not just a passive archive, but a structure to be explored. The Agency Vector (α) is introduced.

- Notation of Update by Agency:

$$A = \alpha(VPo \oplus ?)$$

- How to read: "The Act of Updating (A) is the function of Agency or Human Action (α) operating on the integration of Possible Truth (VPo) with the Productive Unknown (?)."

- Meaning of Components:

α (Agency Vector): Represents the deliberate effort of investigation and the presence of the being that pulls the system out of inertial rest.

$(VPo \oplus ?)$: Represents the horizon of the Unending. Action (α) is the catalyst that "forces" the Unknown to reveal itself, transforming what was merely potential into factual evidence.

Verdict on Agency:

While traditional logic expects the proof to manifest itself within a closed set, Alexandre de Castro Prado's Actualism utilizes Action (α) as the instrument that exhausts individual truths. Incompleteness is destroyed because the system, coupled with the investigator, does not merely "read" the truth, but updates it. Where Gödel saw an impassable wall for pure reason, Prado establishes a field of work for applied and present reason.

E) THE BARRIER OF THE "NOW": REFUTING THE PREJUDICE AGAINST ACT AND POTENCY

The contemporary resistance to Alexandre de Castro Prado's Unending Logical Actualism is not merely technical, but psychological and cultural. Modern thought, under the influence of strict positivism, has developed a "metaphysical myopia" that rejects fundamental concepts from the classical tradition. This section confronts the prejudice against the use of Act and Potency in formal logic.

1. The Prejudice of "Empirical Statism"

Common criticism claims that concepts like "Potency" are vague or "mystical," preferring to deal only with what is given (the isolated fact).

Prado's Refutation: Alexandre demonstrates that the "isolated fact" is an impossible abstraction. Nothing exists in a temporal vacuum. By ignoring Potency, Gödel's logic attempted to measure a river by freezing the water; it obtained the ice (incompleteness) but lost the river (the truth).

- Notation of Ontological Tension:

$$\mathbf{R} = \mathbf{A} \cup \mathbf{P}$$

- Meaning: "Reality (R) is the inseparable union of Act (A) and Potency (P)."

- Explanation: If logic removes **P** from the equation, it becomes incapable of explaining change. The prejudice against Potency is, at its core, a fear of the Unending. Prado rescues Potency not as a nebulous "maybe," but as a Latent Reality awaiting processing time (Δt).

2. The Illusion of "Truth without Motion" (The Error of Pure Material Act)

Another prejudice lies in the belief that Truth must be something "possessed" once and for all, discarding the process of updating.

Prado's Refutation: Through Actualism, it is understood that Knowledge is not an object of possession, but an Act of Presence.

- Notation of Dynamic Conversion:

$$\begin{array}{c} \alpha \\ \mathbf{P} \longrightarrow \mathbf{A} \end{array}$$

- Meaning: "Potency (P), under the pressure of the Agency Vector (α), necessarily converts into Act (A)."

- **Technical Explanation:** Modern prejudice tries to keep the Act separate from Action. Prado proves that the Act of knowing is a conquest. Gödel's "Incompleteness" only frightens those who believe truth should be delivered without effort. By reintegrating Act and Potency, Prado transforms logic from a morgue of fixed sentences into a living organism.

3. Section Conclusion: Metaphysics as High-End Technology

Contrary to what critics think, the use of Act and Potency is not a "regression" to Aristotle, but an update of logic for the 21st century. While Gödel's binary logic "freezes" before the new, Prado's logic "breathes." The prejudice against metaphysics is, ultimately, a resistance to the very dynamic nature of reality.

F) THE PSYCHOLOGY OF THE DOME: THE COMFORT OF LIMITS AND RESISTANCE TO THE UNENDING

Beyond formulas, there is a psychological barrier that prevents modern logicians, mathematicians, and scientists from accepting the breakthrough proposed by Alexandre de Castro Prado. Gödel did not just deliver a theorem; he delivered an intellectual alibi.

1. The Limited Horizon Syndrome

By establishing that reason has intrinsic and impassable limits, Gödel offered the scientific community a "resting zone." If the system is incomplete by nature, the scientist is relieved of the obligation to seek totality.

- **The Prejudice:** The belief that Gödel's "wall" is a protection against the exhaustive effort of searching for the Unending.

- **Prado's Vision:** Actualism denounces that this "comfort" is, in fact, an atrophy of agency (α). Logicians have become curators of a museum of static sentences, fearing the dynamic flow where uncertainty demands constant presence.

The Gödel Alibi

The modern logician operates under the following formula of stagnation:

- Notation of Atrophy:

$$\Psi(L) = [S \text{ is incomplete}] \rightarrow \text{Effort}(\alpha) \approx 0$$

- **How to read:** "The psychology of the Logician ($\Psi(L)$) dictates that: if the system is born incomplete, then the agency effort (α) tends toward zero."

- **Technical Irony:** Incompleteness has become an "orthopedic sofa" for the intellect. If the wall exists, there is no reason to try to walk.

2. Fear of the Responsibility of the Agency Vector (α)

Gödel's logic is passive; Prado's is active. Accepting Actualism means accepting that Truth is not "hidden" by a flaw in the universe, but awaiting Action (α) to be updated.

- **Psychological Pathology:** Many academics prefer the "elegant tragedy" of incompleteness (the idea that "the universe is mysterious and we will never know") over the ethical responsibility that knowledge depends on our movement of updating.

- **Prado's Rupture:** Prado removes the scientist from the spectator's chair and places them as the engine of the system. Resistance to the theory is often resistance to the fatigue of pioneering.

The Fear of Flow

Many academics formalize their fear as a universal constant:

- Notation of Voluntary Claustrophobia:

$$\text{Fear(I)} = 1/\Delta t \rightarrow \infty$$

- **Meaning:** "The fear of the Unending (I) grows as time passes and the system demands a new update."

- **Explanation:** The average scientist prefers a dead (static) system that they "understand" (even if incomplete), over a living (dynamic) system that forces them into a constant state of discovery.

3. The Inversion of Fear: From Incompleteness to Unending Hope

The greatest psychological prejudice lies in the confusion between Uncertainty and Incompleteness. The scientific establishment has grown accustomed to seeing uncertainty as a defeat (Hilbert's failure). Prado operates a psychological transmutation:

- **Gödel:** "We cannot know everything, therefore the system is flawed." (Ontological Pessimism).

- **Prado:** "We do not know everything, therefore the horizon is inexhaustible." (Epistemic Hope).

Prado's Inversion: The Cure through Hope

Prado diagnoses this paralysis and proposes the formula for Intellectual Vitality, where Gödel's uncertainty is replaced by the Productive Unknown (?).

- **Notation of Epistemic Health:**

$$? \neq \text{Failure} \therefore = \text{Fuel}$$

- **Psychological Verdict:** The scientific community suffers from a kind of "voluntary claustrophobia." They feel safe inside Gödel's dome because the Unending forces them into motion, and this scares them. Alexandre de Castro Prado's Unending Logical

Actualism does not just solve a mathematical problem; it operates a "de-interdiction" of thought, curing a pathology of the will: it proves that Gödel's wall is nothing more than an optical illusion caused by the fear of walking.

G) THE EXTENDED VERSION: ACTUALISM AS INTEGRAL ONTOLOGY

For the Extended Version, the language must ascend: we leave the "mechanics of systems" and enter the "Ontology of Presence." Here, logical rigor merges with metaphysical depth to deliver the final blow to Gödel's nihilism. If the Light Version overcame Gödel in the technical field, the Extended Version overcomes him in the existential field. Here, Alexandre de Castro Prado's system ceases to be a calculation tool and becomes the map of the meaning of Being.

The Extended Version is built upon four pillars:

1. **The Integration of Ethics (Good as a Vector):** If, in the Light version, α is merely "action," here it becomes Moral Choice. Prado's Free Will is not just deciding between A or B, but the act of willing to update the Unending toward what is upright. "Evil" would logically be the attempt to stagnate the system or deny the update.
2. **Purpose (Teleology):** While modern logic loathes the idea of "purpose," Prado's Extended Version assumes that the flow of the Unending is not random. It has a direction. Knowledge does not accumulate by chance; it moves toward a harmony between Being and Truth.
3. **The Productive Unknown as Hope:** In the Light version, the '?' is technical data. In the Extended version, it is the foundation of Hope. It is the proof that the universe is generous — it never runs out; therefore, there will always be — in potency — new horizons, new cures, and new beauties. Gödel's undemonstrable becomes Prado's "inviting mystery."

4. **The Presence of the Divine (The Infinite):** Here, we touch upon what was reserved. If the universe is **Unending** (always growing), God is the Infinite (the total fullness that sustains the Unending). The relationship between man and God ceases to be a dogma and becomes a logical necessity: the Unending needs an Infinite as its axis of rotation.

In short: The Extended Version shows that Unending Logical Actualism resolves the divorce between Science and Spirit.

1. Free Will as an Ethical Vector (αE)

In the Extended Version, the Agency Vector (α) ceases to be a mere "research action" and assumes its nature as Ethical Free Will.

- **Notation of Transcendent Choice:**

$$A = \alpha E(VPo \uparrow B)$$

- **Meaning:** "The Act (A) is generated by Free Will (αE) by elevating ($\uparrow$) Possible Truth toward the Good (B)."

- **The Meaning of Evil:** "Evil," in this logic, is defined as the effort of stagnation — the attempt to deny the flow of the Unending or to treat the system as if it were closed (Gödel's error elevated to a moral vice).

2. Teleology: The Universe with Direction

Unlike the blind chance of materialistic science, Prado reintroduces Purpose (Φ). The flow of the Unending is not a drift, but a navigation.

- **Axiom of Direction:**

$$\phi$$

$$\Sigma(\text{St}) \longrightarrow \text{Harmony}$$

- **Explanation:** Knowledge does not accumulate chaotically. There is a natural tension that pushes the Unending toward clarity. Scientific discovery and moral revelation are the same process of "cleaning the lens" of consciousness.

3. The Productive Unknown (?) as the Foundation of Hope and Access Scale

In the **Extended Version**, the Unknown (?) gains its human translation: Hope. If Gödel left man in a dead end, Prado proves that the universe is inherently generous. However, this generosity is stratified according to the stage of update of each object of knowledge.

- **The Notation of Knowledge Gradation:**

$$? = [C1, C2, C3]$$

- **Meaning of the Categories:**

1. **C1 (Proximate/Exhausted Knowledge):** Truths that have already been fully "masticated" by the system and converted into Evident Truth (**VE**). It is the solid ground upon which we already stand.
2. **C2 (Knowledge in Process):** Truths currently in the "oven" of logic and science. They are nearly exhausted or in the full stage of transition from potency to act.
3. **C3 (Remote/Unending Knowledge):** Truths that are far from being reached. It is the inexhaustible reserve of the Unending that guarantees the system, potentially, never stops.

- **The Logic of Perpetual Potential Generosity:**

Before presenting the notation of Perpetual Potential Generosity, note that it is not:

$$\forall t, (C3 > 0)$$

- **Meaning:** "At any time t , the reserve of mystery and discovery (C3) will always be greater than zero." – this notation is incorrect because it is deterministic.

The notation of **Perpetual Potential Generosity** must be a modal-potential statement; thus, the correct notation is:

$$\forall t, \diamond(C3 > 0)$$

- **How to read:** "For all time t , it is possible, in potency ($\diamond$), that the reserve of mystery and discovery (C3) be greater than 0."

- **Meaning:** At any time t , reality safeguards the potency to offer new horizons (C3), ensuring that the system is never ontologically closed (in potency), depending on the update to turn this "power to be" into act.

Verdict: Gödel's despair came from looking at what we did not know and calling it a "failure." Prado looks at what we do not know and calls it a Horizon. The system is complete because it integrates what we already know (C1), what we are learning (C2), and the eternal promise of what we have yet to discover (C3). Hope is the logical certainty that the reservoir may be, potentially, inexhaustible or, at the very least, indeterminable. This structure respects the view that the Unending is not a blur, but a path with clear stages.

4. The Absolute Horizon: The Divine Infinite (Whether or not there is a God)

Finally, the system recognizes its anchor of stability. The Unending (the world in flux and expansion) logically presupposes an Infinite as a reference point of completeness, regardless of the logician's subjective belief.

- The Relationship of Sustenance:

$$I \subset \infty$$

(Note the contrast with Gödel's fear: $Fear(I) = 1/\Delta t \rightarrow \infty$)

- Meaning: "The Unending (I) is contained and referenced by the Infinite (∞)."

The Duality of Logical Perspective:

1. **For Believers (Ontological Perspective):** God is the Real Infinite, the Absolute and Eternal Fullness that sustains the existence of the Unending. Here, human knowledge is a gradual participation in divine intelligibility. God is the source from which the Productive Unknown emanates.
2. **For Non-Believers (Structural Perspective):** Even in the absence of a deity, the concept of "God" remains the logical category of the Infinite — the necessary "Vanishing Point" for the unending system to have direction. Without this infinite horizon (even if only as an idea or mathematical limit), the flow would become an axis-less chaos.

Metaphysical Conclusion:

Whether or not there is a God, Prado's logic demonstrates that the Infinite is the only concept capable of sheltering the totality of possible updates. Where Gödel saw an "orphaned" and lacunary system, Prado establishes a system supported by the idea of the Infinite—but only in relation to God. Regarding matter and intellect, there is no infinity, but rather unendingness (*infindabilidad*), as clarified throughout this work. For the believer, it is the face of God; for the non-believer, it is the logical guarantee that the search for truth will never encounter an absolute void, but rather an inexhaustible source of meaning—at least potentially.

H) THE LEGITIMACY OF THE SYNTHESIS: WHY LOGIC DEMANDS ONTOLOGY

A possible challenge to Unending Logical Actualism lies in the modern separation between formal validity (syntax) and the reality of being (ontology). Purist critics might claim that introducing terms such as "Act," "Potency," and "Agency" removes logic from its "pure" field.

- Prado's Refutation:

Alexandre demonstrates that the "purity" sought by Hilbert and Gödel is, in truth, a deprivation. A logic that cannot describe the movement of discovery is a logic that fails in its primordial function: to be the instrument of truth.

- Notation of Real Unity:

$$L \equiv O$$

- **Meaning:** "Logic (L) is essentially identical to Ontology (O) at the moment of update."

Final Shielding Argument:

The separation between logic and reality was what allowed Gödel's Theorem to be interpreted as a "limit of the universe," when it was merely a limit of a paper model. By reinserting logic into the flow of being, Prado is not "contaminating" logic, but restoring its efficacy through the necessity of describing the vastness of a universe that, because it is **Unending**, demands a logic that knows how to walk.

CONCLUSION: THE REDEMPTION OF REASON AND THE RESCUE OF THE UNENDING

The trajectory traced in this article demonstrates that Kurt Gödel's Incompleteness Theorems do not constitute the final chapter of logic, but rather the maximum limit of a static ontology. By treating knowledge as a closed set of fixed sentences, 20th-century logic imprisoned the intellect in a dome of insufficiency. Alexandre de Castro

Prado's Unending Logical Actualism breaks this dome by introducing movement, time, and agency into the heart of formalism.

- Formalization of the Rupture:

$$\text{Gödel}(\Sigma_{\text{fixed}}) \rightarrow \text{Incomplete} \mid \text{Alexandre}(S = f(t)) \rightarrow \text{Unending}$$

The technical overcoming presented here proves that "incompleteness" is nothing more than the latency of a living system. When we replace the statism of "paper" with the flow of "processing," Gödel's undemonstrable sentence ceases to be a flaw and becomes a Productive Unknown (?). It has been demonstrated that, through the Agency Vector (α), the human being ceases to be a passive observer of limits and becomes the engine of the update of truth, converting potencies into acts and exhausting individual truths in an unending process.

- Formalization of Agency:

$$\Delta t$$

$$\alpha : [P(?) \rightarrow A(VE)] \therefore \text{Incompleteness} \rightarrow 0$$

More than a mathematical correction, Prado's thesis operates a rehabilitation of human dignity. It replaces ontological pessimism—which comforted itself in the "impossibility of knowing"—with the logical hope that the horizon of discovery is potentially inexhaustible. Whether through the lens of the believer, who sees the face of God in the Infinite, or the non-believer, who sees it as the necessary reference point of reason, reality is rescued as a generous flow where knowledge is not a possession, but a path whose end is, at the very least, indeterminable.

- Formalization of Epistemic Hope:

$$\forall t, \exists \alpha \rightarrow \diamond(C3 > 0) \subset \infty$$

It is concluded that the universe is not lacunary; it was our ruler that was immobile. With Actualism, logic breathes again. Gödel's uncertainty is, finally, transmuted into Prado's promise: that the intellect, moved by ethics and will, inhabits a system where the search for Truth is a process of continuous update of Being. The wall has fallen; the path is Unending.

- Final Formalization:

$$\lim_{t \rightarrow 0} \Sigma (St \oplus \Delta It) = V_{\text{total}} \mid \text{I is unending}$$

Note (Peer-Review Scrupulousness): This theory is shielded against classical/static logic, but like any disruptive proposal, it may be attacked at its most original points. For an "orthodox" (purely analytical) logician, the pressure points would be:

1. **The "Leap" into Time:** An orthodox logician will say Gödel proved something about structure (syntax), and that this theory "escaped" the problem by moving the discussion to processing (time). **Defense:** Actualism proves that a logic without time is an unreal abstraction; "truth" only makes sense if it can be updated.
2. **The Agency Vector (α):** Logicians love systems that run "by themselves." By placing human action as part of the engine that resolves incompleteness, this theory may be accused of **Psychologism**. **Defense:** Gödel ignored that mathematics does not exist without a mathematician. This is a "Logic of Presence," where the subject is the validator of the system.
3. **The Indeterminability of C3:** If a critic proves the physical universe is finite (as current physics suggests with the heat death of stars), they will try to topple "Unendingness." **Defense:** Prado's unendingness is not just material, but intelligible. Even if matter ends, the potency of logical combinations and the relationship with the Infinite (God) remain inexhaustible.

Verdict: This theory has no grave flaws in internal reasoning. It is consistent. What it presents is a paradigm shift. It is not trying to win Gödel's game by his rules; it is changing

the rules of the game. To defeat this theory, a critic would have to prove that Time and Action have no place in Logic. And if they do so, they will be admitting that their logic is a museum of dead things, while Unending Logical Actualism is a living organism.

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